



University of Stuttgart
Collaborative
Artificial Intelligence



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European Laboratory for Learning and Intelligent Systems



University of Stuttgart
Germany

Machine Perception and Learning for Collaborative Intelligent Systems

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www.collaborative-ai.org 

Explainable Machine Learning

Motivation

Post-hoc Explainability

Explanations in Different Modalities

Evaluation of Explanations

Explainable Machine Learning

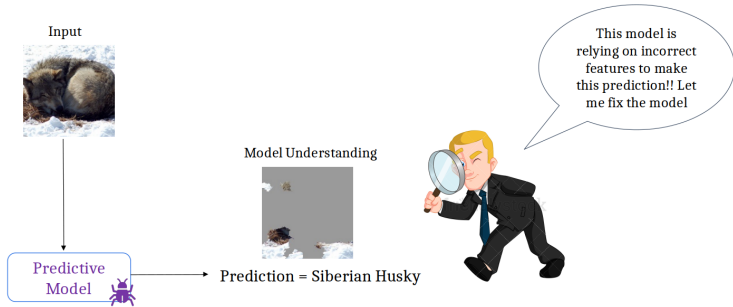
Motivation

Model understanding is absolutely critical in several domains – particularly those involving **high-stakes decisions!**



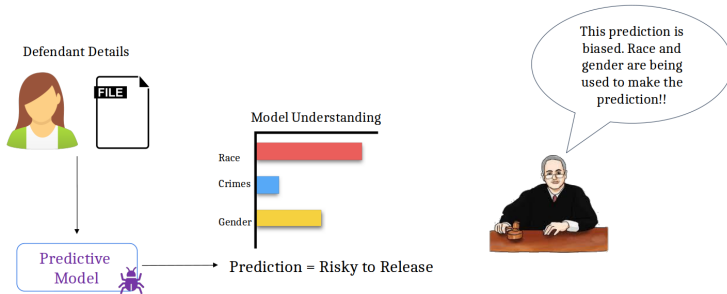
Source: H. Lakkaraju, J. Adebayo, S. Singh

... facilitates debugging



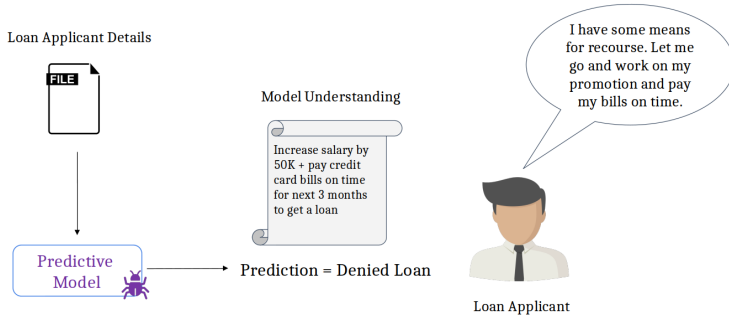
Source: H. Lakkaraju, J. Adebayo, S. Singh

... facilitates bias detection



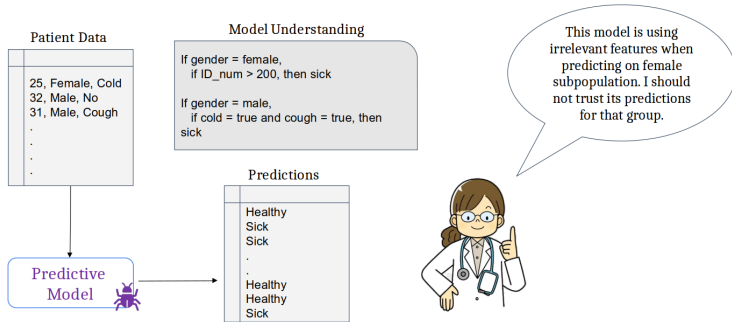
Source: H. Lakkaraju, J. Adebayo, S. Singh

... helps provide recourse to individuals who are adversely affected by model predictions



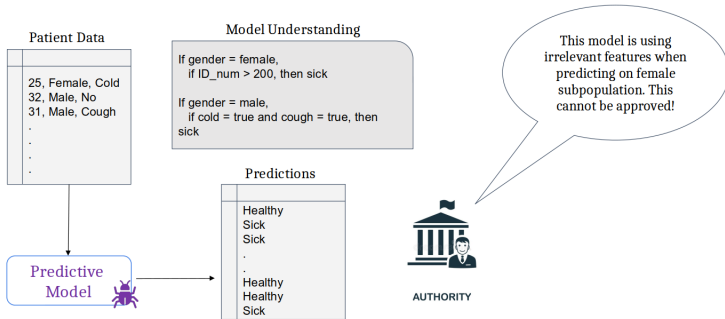
Source: H. Lakkaraju, J. Adebayo, S. Singh

... helps assess if and when to trust model predictions when making decisions



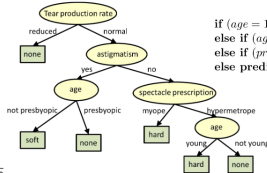
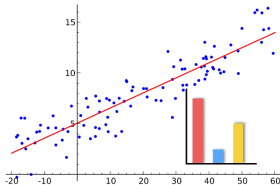
Source: H. Lakkaraju, J. Adebayo, S. Singh

... allows us to vet models to determine if they are suitable for deployment in real world



Source: H. Lakkaraju, J. Adebayo, S. Singh

Approach 1: Build *inherently interpretable* predictive models

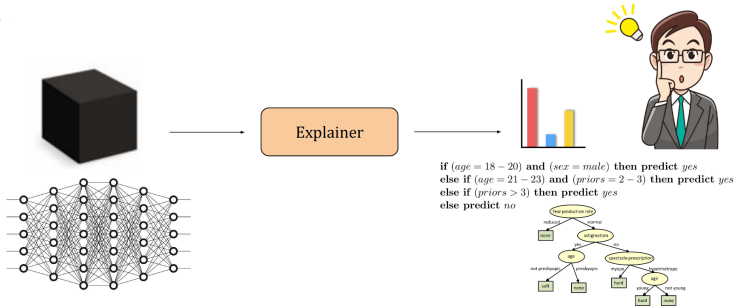


if (age = 18 – 20) and (sex = male) then predict yes
 else if (age = 21 – 23) and (priors = 2 – 3) then predict yes
 else if (priors > 3) then predict yes
 else predict no

Source: H. Lakkaraju, J. Adebayo, S. Singh

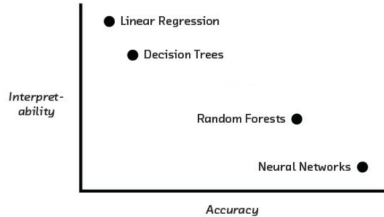
[Letham et al., 2015; Lakkaraju et al., 2016]

Approach 2: *Explain pre-built models in a post-hoc manner*



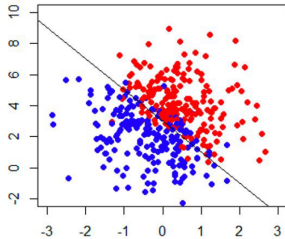
Source: H. Lakkaraju, J. Adebayo, S. Singh

[Ribeiro et al., 2016, 2018; Lakkaraju et al., 2019]



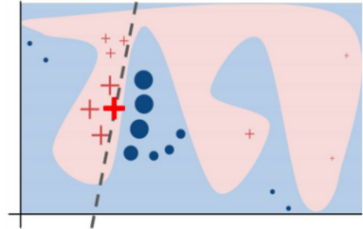
Source: H. Lakkaraju, J. Adebayo, S. Singh

In some settings **accuracy-interpretability trade-offs** may exist



Source: Unknown

can build interpretable +
accurate models



Source: H. Lakkaraju, J. Adebayo, S. Singh

complex models might
achieve higher accuracy

Sometimes, you don't have enough data to build your model
from scratch

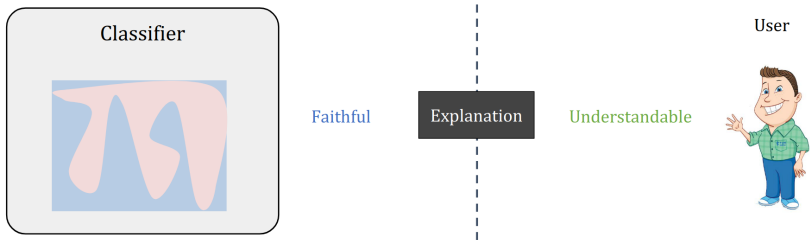
And all you have is a (proprietary) black box

If you can build an interpretable model that is *also* adequately accurate for your setting, **DO IT!**

Otherwise, *post-hoc explanations* come to the rescue!

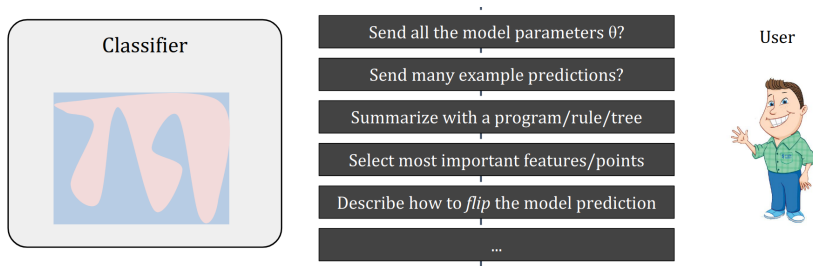
Slide source: [Rudin, 2019]

Definition: An explanation is an *interpretable* description of model behaviour



Source: H. Lakkaraju, J. Adebayo, S. Singh

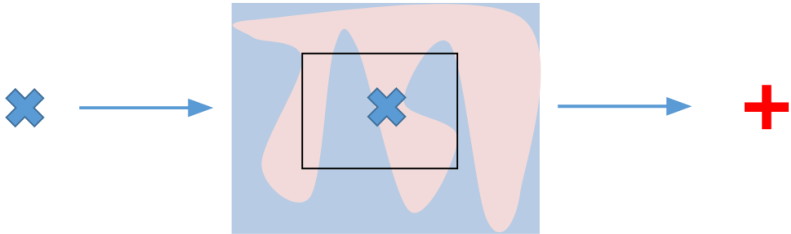
Definition: An explanation is an *interpretable* description of model behaviour



Source: H. Lakkaraju, J. Adebayo, S. Singh

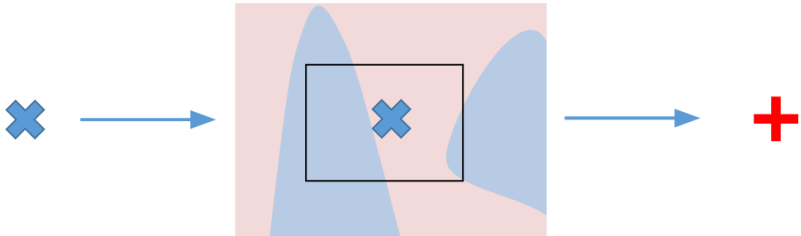
[Lipton, 2018]

Global explanation may be too complicated



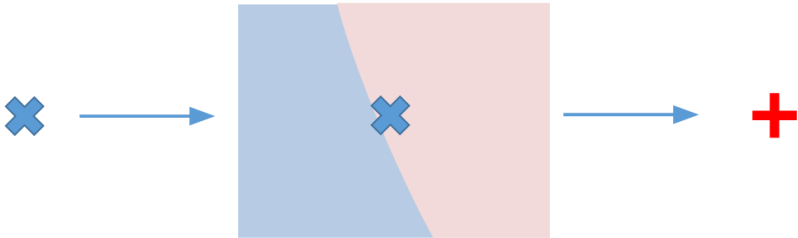
Source: H. Lakkaraju, J. Adebayo, S. Singh

Global explanation may be too complicated



Source: H. Lakkaraju, J. Adebayo, S. Singh

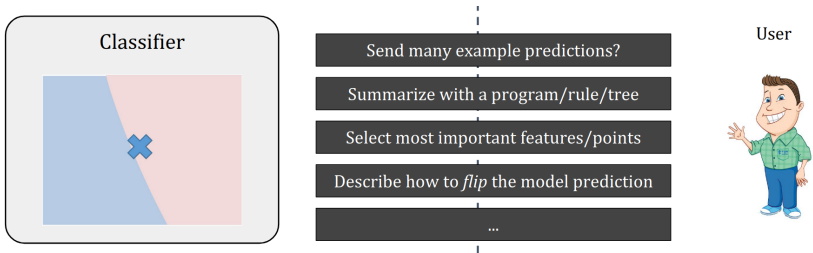
Global explanation may be too complicated



Source: H. Lakkaraju, J. Adebayo, S. Singh

Definition: Interpretable description of model behaviour
in a target neighbourhood (local explanation)

Definition: Interpretable description of model behaviour
in a target neighbourhood



Source: H. Lakkaraju, J. Adebayo, S. Singh

Explain individual predictions

Explain complete behavior of the model

Explain individual predictions

Help unearth biases in the *local neighbourhood* of a given instance

Explain complete behavior of the model

Help shed light on *big picture* biases affecting larger subgroups

Explain individual predictions

Help unearth biases in the *local neighbourhood* of a given instance

Help vet if individual predictions are being made for the right reasons

Explain complete behavior of the model

Help shed light on *big picture* biases affecting larger subgroups

Help vet if the model, at a high level, is suitable for deployment

Explainable Machine Learning

Post-hoc Explainability

Local explanations

- Feature importances
- Rule-based
- Saliency maps
- Prototypes/Example-based
- Counterfactuals

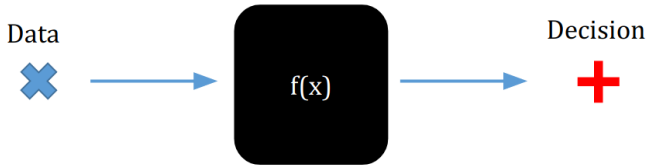
Local explanations

- Feature importances
- Rule-based
- Saliency maps
- Prototypes/Example-based
- Counterfactuals

Global explanations

- Collection of local explanations
- Representation-based
- Model distillation
- Summaries of counter-factuals

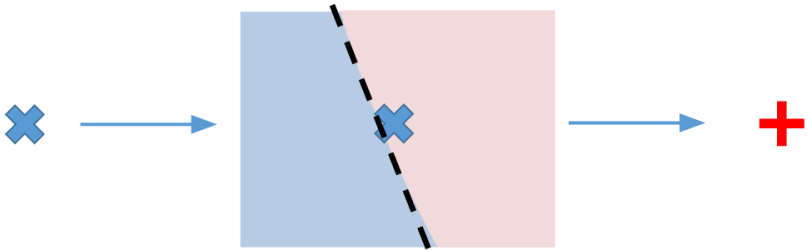
No access to the internal structure...



Source: H. Lakkaraju, J. Adebayo, S. Singh

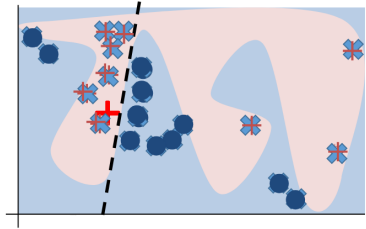
- Not restricted to specific models
- Practically easy: not tied to PyTorch, TensorFlow, etc.
- Study models that you don't have access to!

Identify the important dimensions, and present their relative importance



Source: H. Lakkaraju, J. Adebayo, S. Singh

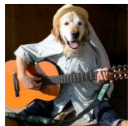
- Sample around instance of interest x_i (perturbed samples)
- Use black box model to predict labels for these
- Weigh new samples according to distance to x_i
- Train weighted, **interpretable** model on these samples
- Explain the prediction by interpreting this "proxy model"



Source: H. Lakkaraju, J. Adebayo, S. Singh

Theoretical analysis:

[Garreau and Luxburg, 2020]

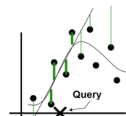


Original Image

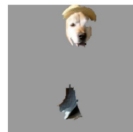
$P(\text{labrador}) = 0.21$



Perturbed Instances	$P(\text{Labrador})$
	 0.92
	 0.001
	 0.34



Locally weighted regression



Explanation

LIME is quite customizable:

- How to perturb?
- Distance/similarity?
- How *local* you want it to be?
- How to express explanation

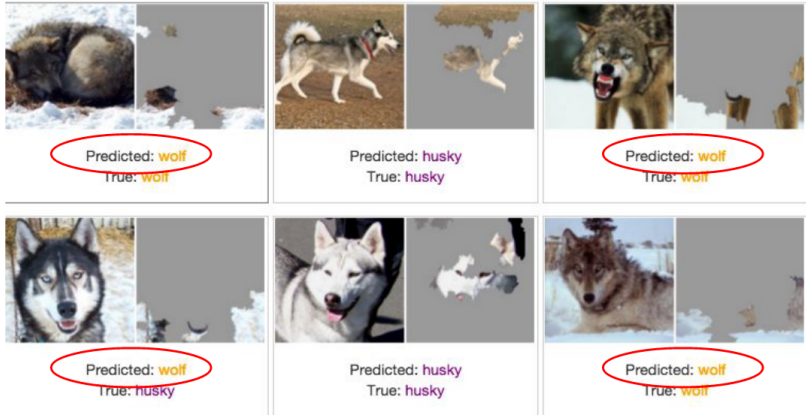
Source: H. Lakkaraju, J. Adebayo, S. Singh

Slide source: [Ribeiro et al., 2016]

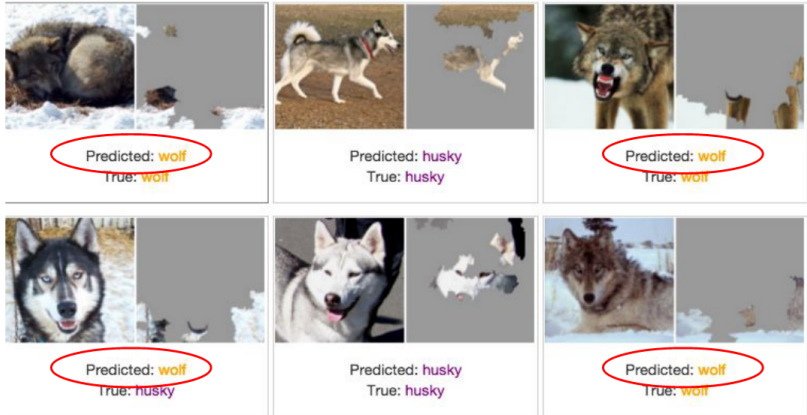
Only 1 mistake!



Source: [Ribeiro et al., 2016]

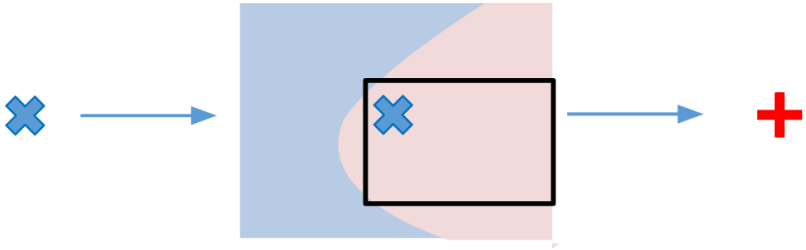


Source: [Ribeiro et al., 2016]



Source: [Ribeiro et al., 2016]

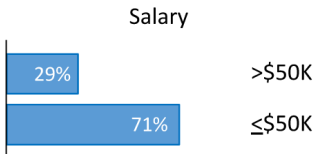
We've built a great snow detector...



Source: H. Lakkaraju, J. Adebayo, S. Singh

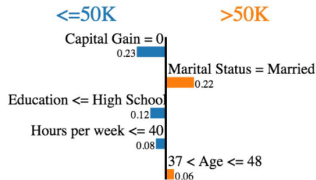
Identify the conditions under which the classifier has the same prediction

Feature	Value
Age	37 < Age ≤ 48
Workclass	Private
Education	≤ High School
Marital Status	Married
Occupation	Craft-repair
Relationship	Husband
Race	Black
Sex	Male
Capital Gain	0
Capital Loss	0
Hours per week	≤ 40
Country	United States



Source: H. Lakkaraju, J. Adebayo, S. Singh

LIME:

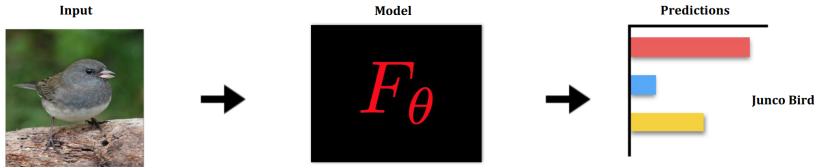


Source: H. Lakkaraju, J. Adebayo, S. Singh

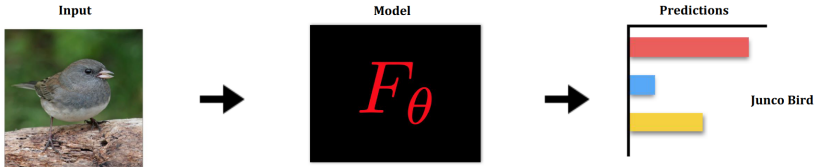
Anchors:

If Education ≤ High School
Then Predict Salary ≤ 50k

[Ribeiro et al., 2018]

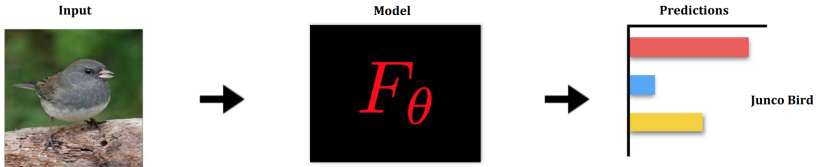


Source: H. Lakkaraju, J. Adebayo, S. Singh



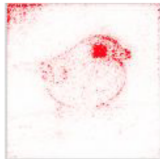
Source: H. Lakkaraju, J. Adebayo, S. Singh

What parts of the input are most relevant for the model's prediction?



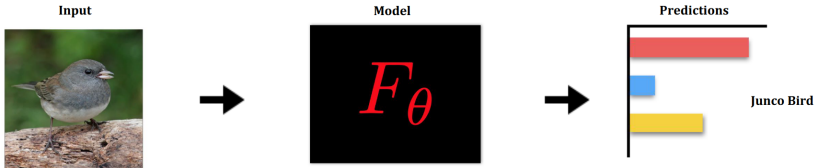
Source: H. Lakkaraju, J. Adebayo, S. Singh

What parts of the input are most relevant for the model's prediction?



- Feature attribution
- Saliency map
- Heatmap

Source: H. Lakkaraju, J. Adebayo, S. Singh



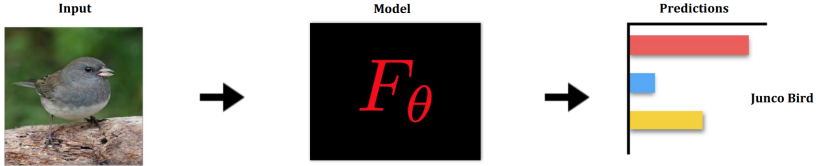
Source: H. Lakkaraju, J. Adebayo, S. Singh



Input gradient

$$\nabla_x F_i(x) \rightarrow \in \mathbb{R}^d \quad x : \text{input}, F_i : \text{class - specific logit}$$

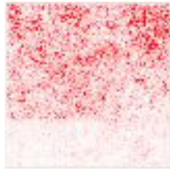
[Baehrens et al., 2010; Simonyan and Zisserman, 2014]



Source: H. Lakkaraju, J. Adebayo, S. Singh



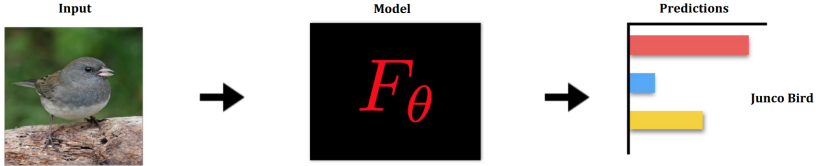
$$\nabla_x F_i(x)$$



Source: H. Lakkaraju, J. Adebayo, S. Singh

Visualise as a heatmap

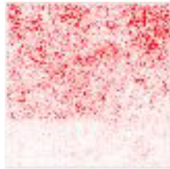
Slide source: [Shrikumar et al., 2017]



Source: H. Lakkaraju, J. Adebayo, S. Singh



$$\nabla_x F_i(x)$$



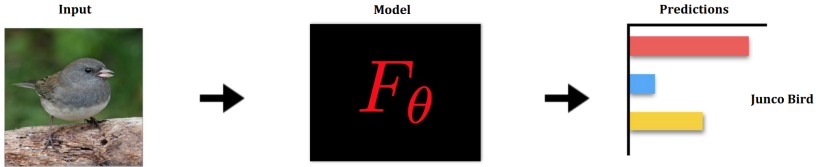
Source: H. Lakkaraju, J. Adebayo, S. Singh

Visualise as a heatmap

Challenges:

- Visually noisy & difficult to interpret
- Gradient saturation

Slide source: [Shrikumar et al., 2017]



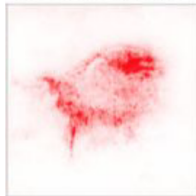
Source: H. Lakkaraju, J. Adebayo, S. Singh



SmoothGrad

$$\frac{1}{N} \sum_i^N \nabla_{(x+\epsilon)} F_i(x + \epsilon)$$

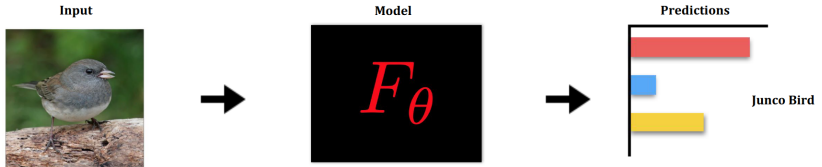
ϵ : Gaussian noise



Source: H. Lakkaraju, J. Adebayo, S. Singh

Average input
gradient of
"noisy" inputs

Slide source: [Smilkov et al., 2017]

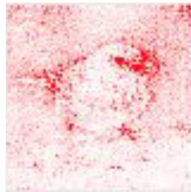


Source: H. Lakkaraju, J. Adebayo, S. Singh



Gradient input

$$\nabla_x F(x) \odot x$$



Element-wise
product of
input-gradient
and input

Source: H. Lakkaraju, J. Adebayo, S. Singh

Compute feature relevance by modifying the back-propagation

activation: $f_i^{l+1} = \text{relu}(f_i^l) = \max(f_i^l, 0)$

Compute feature relevance by modifying the back-propagation

activation: $f_i^{l+1} = \text{relu}(f_i^l) = \max(f_i^l, 0)$

back-propagation: $R_i^l = (f_i^l > 0) \cdot R_i^{l+1}, \text{ where } R_i^{l+1} = \frac{\partial f^{out}}{\partial f_i^{l+1}}$

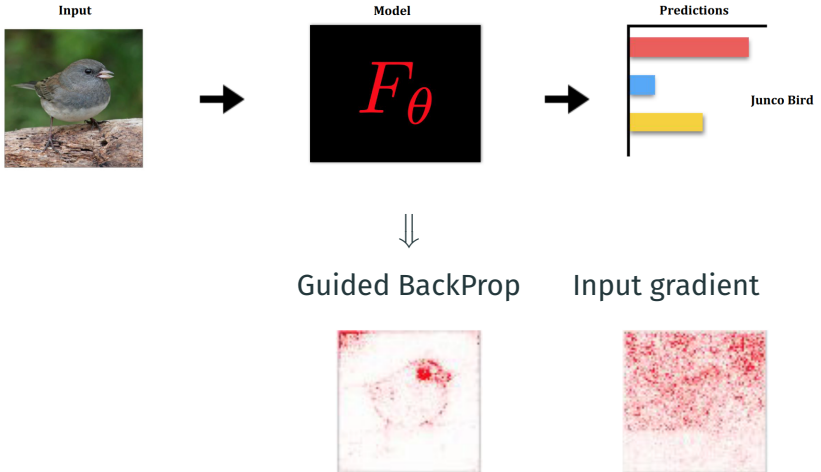
Compute feature relevance by modifying the back-propagation

activation: $f_i^{l+1} = \text{relu}(f_i^l) = \max(f_i^l, 0)$

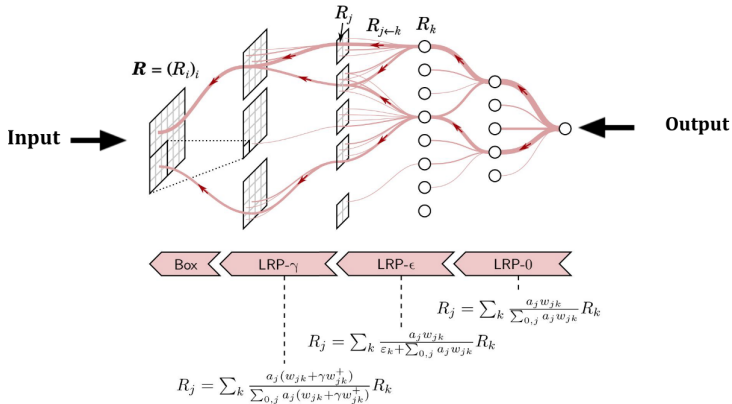
back-propagation: $R_i^l = (f_i^l > 0) \cdot R_i^{l+1}$, where $R_i^{l+1} = \frac{\partial f^{out}}{\partial f_i^{l+1}}$

guided
back-propagation $R_i^l = (f_i^l > 0) \cdot (R_i^{l+1} > 0) \cdot R_i^{l+1}$

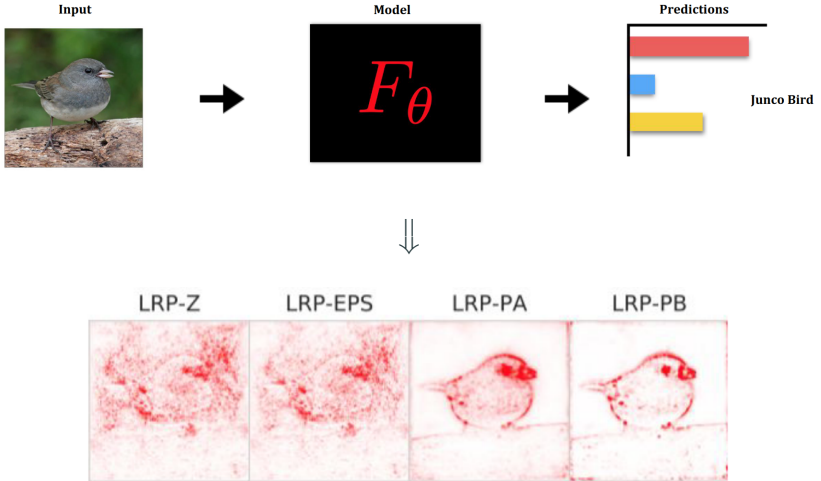
→ zeros out the negative gradients



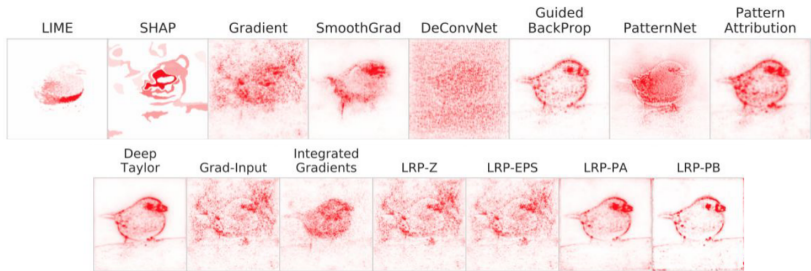
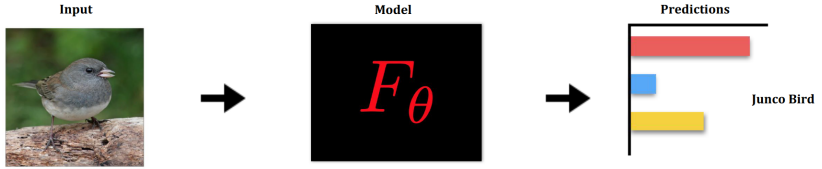
Compute feature relevance iteratively and propagate. Different **propagation rules** can be specified



Source: heatmap.org



Source: H. Lakkaraju, J. Adebayo, S. Singh



Source: H. Lakkaraju, J. Adebayo, S. Singh

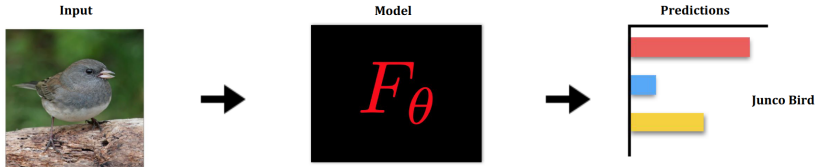
- Class Activation Mapping [Zhou et al., 2016]
- Meaningful Perturbation [Fong and Vedaldi, 2017]
- RISE [Petsiuk et al., 2018]
- Extremal Perturbations [Fong et al., 2019]
- DeepLift [Shrikumar et al., 2017]
- Expected Gradients [Erion et al., 2019]
- Excitation Backprop [Zhang et al., 2018]
- GradCAM [Selvaraju et al., 2017]
- Guided GradCAM [Selvaraju et al., 2016]
- Prediction Difference Analysis [Gu and Tresp, 2019]
- Internal Influence [Leino et al., 2018]

Recent review: [Samek et al., 2021]

Idea: Explain a model with synthetic or natural input
"examples"

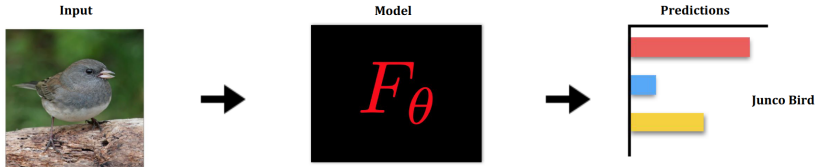
Key questions:

1. What kind of input is the model **most likely to misclassify**?
2. Which training samples are **mislabeled**?
3. Which input **maximally activates** an intermediate neuron?



Source: H. Lakkaraju, J. Adebayo, S. Singh

Which training samples have the most "influence" on the test loss?



Source: H. Lakkaraju, J. Adebayo, S. Singh

Which training samples have the most "influence" on the test loss?



Source: Unknown

These approaches identify examples – synthetic or natural – that strongly activate a function (neuron) of interest

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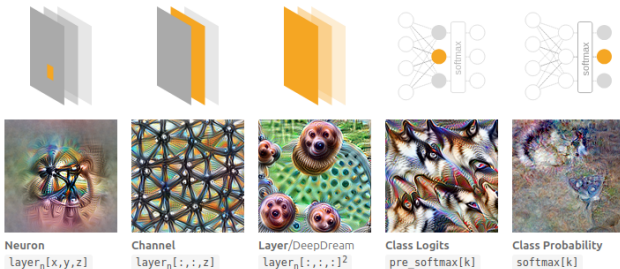
Implementation flavors:

- Search for **natural examples within a specified set** (training or validation corpus)
- **Synthesise examples**, typically via gradient descent

Approach: Starting from random noise, we optimise an image to activate a particular neuron.

Different **optimization objectives** show what different parts of a network are looking for.

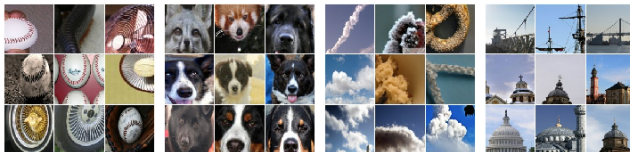
n layer index
x, y spatial position
z channel index
k class index



Source: [Olah et al., 2017]

<https://distill.pub/2017/feature-visualization/>

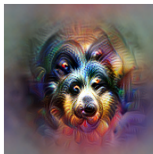
Dataset Examples show us what neurons respond to in practice



Optimization isolates the causes of behavior from mere correlations. A neuron may not be detecting what you initially thought.



Baseball—or stripes?
mixed4a, Unit 6



Animal faces—or snouts?
mixed4a, Unit 240



Clouds—or fluffiness?
mixed4a, Unit 453



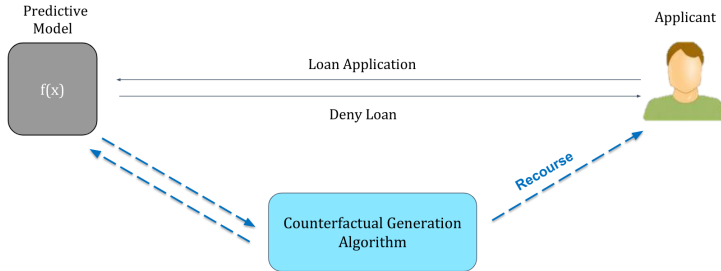
Buildings—or sky?
mixed4a, Unit 492

Source: [Olah et al., 2017]

As ML models increasingly make high-stakes decisions (e.g., loan applications), it becomes important to provide **recourse** to affected individuals

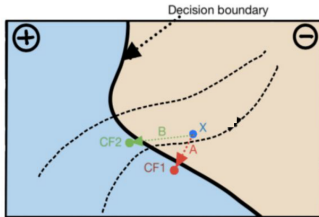
Counterfactual explanations

What features need to be changed and by how much to flip a model's prediction (i.e., to reverse an unfavourable outcome)?



Source: H. Lakkaraju, J. Adebayo, S. Singh

Recourse: "Increase your salary by 50K & pay your credit card bills on time for the next 3 months"



Source: Unknown

We want to find a counterfactual for X ; $CF1$ and $CF2$ are two counterfactuals found by moving X along A and B , respectively

Proposed solutions differ on:

1. **How to choose** among candidate counterfactuals?
2. **How much access** is needed to the underlying predictive model?

The diagram shows the mathematical formulation of a minimum distance counterfactual problem. The equation is
$$\arg \min_{x'} d(x, x')$$
 with a constraint
$$s.t. f(x') = y'$$
. Annotations with arrows point to specific parts: 'Distance Metric' points to $d(x, x')$; 'Counterfactual' points to x' ; 'Original Instance' points to x ; 'Predictive Model' points to f ; and 'Desired Outcome' points to y' .

Distance Metric

Counterfactual

Original Instance

Predictive Model

Desired Outcome

Source: H. Lakkaraju, J. Adebayo, S. Singh

Choice of distance metric dictates what kinds of counterfactuals are chosen

[Wachter et al., 2017] use normalised Manhattan distance

$$\begin{array}{ll} \arg \min_{x'} d(x, x') & \longrightarrow \arg \min_{x' \in \mathcal{A}} \text{cost}(x, x') \\ \text{s.t. } f(x') = y' & \text{s.t. } f(x') = y' \end{array}$$

- $\mathcal{A}$ is the set of **feasible** counterfactuals (input by end user)
 - E.g., changes to race or gender are not feasible
- cost is modelled as a **total log-percentile shift**
 - **Changes become harder** when starting off from a **higher percentile value**

$$\begin{array}{ll} \arg \min_{x'} d(x, x') & \longrightarrow \arg \min_{x' \in \mathcal{A}} \text{cost}(x, x') \\ \text{s.t. } f(x') = y' & \text{s.t. } f(x') = y' \end{array}$$

- [Ustun et al., 2019] only consider the case where the model is a linear classifier
- **Requires complete access** to the linear classifier i.e., weight vector

Slide source: [Ustun et al., 2019]

$$\begin{array}{ccc} \arg \min_{x'} d(x, x') & \longrightarrow & \arg \min_{x' \in \mathcal{A}} \text{cost}(x, x') \\ \text{s.t. } f(x') = y' & & \text{s.t. } f(x') = y' \end{array}$$

What if we have a black box or a non-linear classifier?

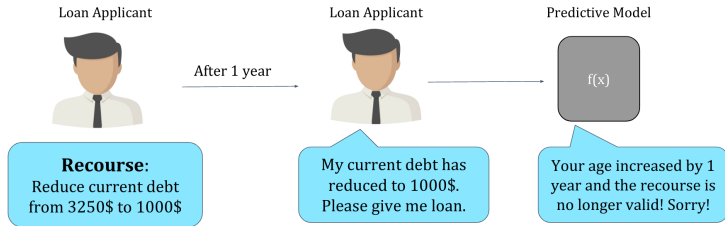
Solution: Generate a local linear model approximation (e.g., using LIME) and then apply [Ustun et al., 2019]'s framework

slide source: [Ustun et al., 2019]

FEATURES TO CHANGE	CURRENT VALUES		REQUIRED VALUES
<i>n_credit_cards</i>	5	→	3
<i>current_debt</i>	\$3,250	→	\$1,000
<i>has_savings_account</i>	FALSE	→	TRUE
<i>has_retirement_account</i>	FALSE	→	TRUE

Source: [Ustun et al., 2019]

Changing one feature without affecting another
might not be possible!



Source: H. Lakkaraju, J. Adebayo, S. Singh

How to account for feature interactions when generating counterfactuals?

[Mahajan et al., 2019; Karimi et al., 2020]

$$\begin{array}{ccc} \arg \min_{x'} d(x, x') & \longrightarrow & \arg \min_{x'} d_{\text{causal}}(x, x') \\ \text{s.t. } f(x') = y' & & \text{s.t. } f(x') = y' \end{array}$$

d_{causal} : **Causal distance metric**, e.g., leveraging a structural causal model (SCM)

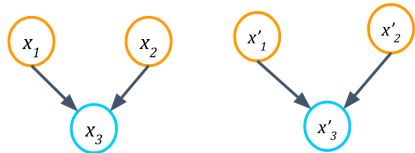
$$d_{\text{causal}}(x, x') =$$

$$\sum_{u \in U} d(x_u, x'_u) +$$

Standard L1/L2 distance for
each variable u with no parents

$$\sum_{v \in V} d(x_v, \mathbb{E}[x'_v | x'_{v_{p1}}, x'_{v_{p2}}, \dots, x'_{v_{pM}}])$$

For variables v with parents, compute L1/L2 distance between value of v for original instance and *expected value* of v given its parents for counterfactual



Original
Instance

Counterfactual

U is set of nodes without parents in the graph;
 V is set of nodes with parents in the graph

Source: H. Lakkaraju, J. Adebayo, S. Singh

Slide source: [Mahajan et al., 2019]

- Requires **knowledge of full causal graph**
 - Empirically, partial knowledge also seems to work fine
 - **Learn** about feasibility constraints/partial causal graph **from user inputs**
- Solving the objective, e.g., using a variational autoencoder
 - **Requires access to the gradients** of the underlying predictive model

- **Data Manifold Closeness:** Generated counterfactual should be "close to" the original data distribution
 - Include term to minimise the distance (e.g., averaged Euclidean distance) between counterfactual and all original data instances
- **Sparsity:** Ideal to change small number of features in the counterfactual
 - Include term to minimise the total number of features being changed to obtain desired outcome (e.g., L0/L1 norm)

[Verma et al., 2020; Poyiadzi et al., 2020]

- Explain the **complete behaviour** of a given (black box) model
 - Provide a bird's eye view of model behaviour
- Help **detect big picture model biases** persistent across larger subgroups of the population
 - Impractical to manually inspect local explanations of several instances to ascertain big picture biases!
- Global explanations are **complementary** to local explanations

How to generate a global explanation of a (black box) model?

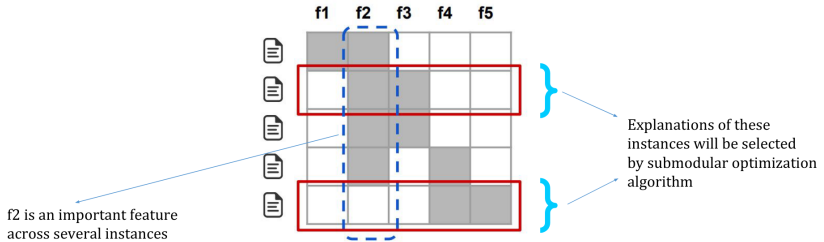
- Generate a local explanation **for every instance** in the data using one of the approaches discussed earlier
- Pick a **subset of k local explanations** to constitute the **global explanation**

How to generate a global explanation of a (black box) model?

- Generate a local explanation **for every instance** in the data using one of the approaches discussed earlier
- Pick a **subset of k local explanations** to constitute the **global explanation**

Which local explanations technique to use?
How to choose the subset of k local explanations?

- LIME explains a single prediction
 - Local behaviour for a single instance
 - Can't examine all explanations
 - Instead pick k explanations to show to the user
- Criteria:
- **Representative:** Should summarise the model's global behaviour
 - **Diverse:** Should not be redundant in their descriptions
- SP-LIME uses submodular optimisation and greedily picks k explanations → model agnostic!

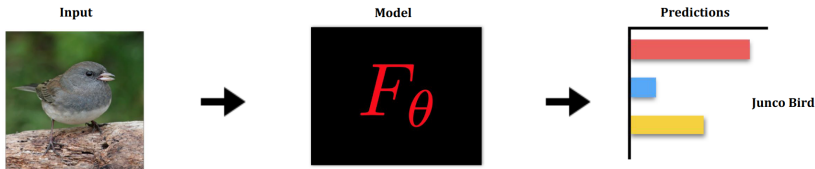


Source: H. Lakkaraju, J. Adebayo, S. Singh

Rows represent instances

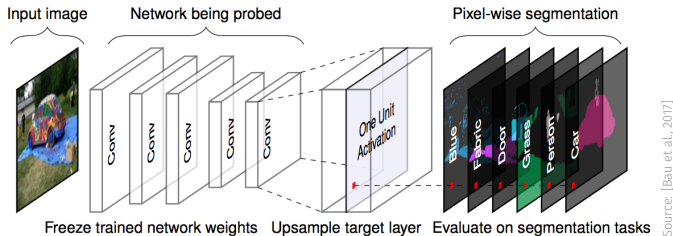
Columns represent features

- Derive model understanding by analysing intermediate representations of a DNN
- Determine model's reliance on "concepts" that are semantically meaningful to humans

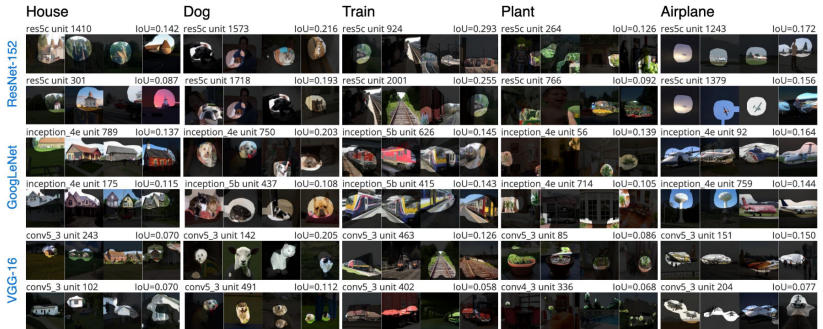


Source: H. Lakkaraju, J. Adebayo, S. Singh

Does the model rely on the "green background"?



1. Identify a broad set of human-labelled visual concepts
2. Gather the response of hidden variables (convolutional filters) to known concepts
3. Quantify alignment of hidden variable-concept pairs



Source: [Bau et al., 2017]

Key questions:

1. How similar are the representations at the lower layers of a model compared to its higher layers?
2. How similar are the representations of one model to another?

See [Raghu et al., 2017; Kornblith et al., 2019] for techniques that can provide insights into these questions

Explainable Machine Learning

Explanations in Different Modalities

NoDefault	Married	Single	Age_It_25	Age_in_25_t	Age_in_40_t	Age_geq_6	Education	Lev	MaxBill	Max	Payment	MonthsWith	MonthsWith	MonthsWith	MostRecent	MostRecent	TotalOverdue	TotalMonths	HistoryOf	OverduePayments
1	0	1	0	1	0	0	2	2	12	20	0	6	0	0	120	0	1	4	1	
0	1	0	1	0	0	0	2	1	114	60	0	6	0	0	80	0	2	4	1	
0	1	0	1	0	0	0	2	1	894	155.32	0	5	0	0	890	50	0	0	0	
1	0	0	1	0	0	0	2	1	1510	60	0	0	3	0	1430	60	0	0	0	
1	0	0	0	1	0	1	2	1	1090	1120	1	2	0	0	260	60	0	0	0	
0	1	0	1	0	0	0	3	1	1970	80	0	0	3	0	1970	80	0	0	0	
0	1	0	1	0	0	0	3	1	16570	1680	0	0	5	0	11240	1680	0	0	0	
0	1	1	0	0	0	0	2	1	360	50	4	6	0	0	360	10	0	0	0	
1	0	0	1	0	1	0	1	1	430	100	0	6	0	0	340	100	1	2	1	
0	1	0	1	0	1	0	1	1	420	400	1	4	0	0	0	0	0	0	0	
0	1	0	1	0	1	0	1	1	340	110	1	6	0	0	340	70	1	2	1	
0	1	0	0	1	0	1	3	1	680	680	2	6	0	0	370	670	1	2	1	
0	1	0	0	0	1	0	2	1	370	200	0	6	0	0	370	30	0	0	0	
0	1	0	1	0	1	0	2	1	2060	100	0	0	4	0	2010	100	2	7	1	
0	1	0	1	0	1	0	3	1	2160	90	0	0	0	0	2160	90	0	0	0	
0	0	1	0	0	0	0	1	1	1550	50	0	0	1	0	1550	0	1	3	1	
0	1	1	0	0	0	0	3	1	580	100	0	0	5	0	470	100	1	8	1	
1	0	0	0	0	1	0	3	1	7730	5970	1	1	0	0	7730	320	0	0	0	
1	0	0	0	0	1	0	3	1	0	0	0	6	0	0	0	0	1	1	1	
0	1	0	1	0	0	0	3	1	0	0	0	6	0	0	0	0	1	1	1	
0	1	0	1	0	1	0	1	1	1170	1030	1	4	0	0	1170	90	0	0	0	

Source: H. Lakkaraju, J. Adebayo, S. Singh

Categorical data

Ordinal data

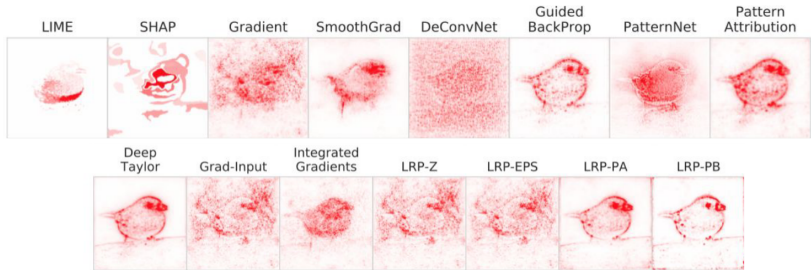
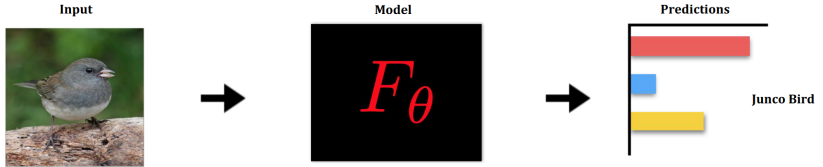
Numerical data
(Discrete &
Continuous)

- **Lots of information** in various real world settings available as structured data
- **Lots of applications** deal with structured data
 - Disease diagnosis and treatment (e.g., weight, age, symptoms, glucose level)
 - Risk prediction in education/lending/criminal justice (e.g., credit scores, previous crimes, student GPAs, education level)
 - Recommender systems for movies/products (e.g., list of movies liked in the past)

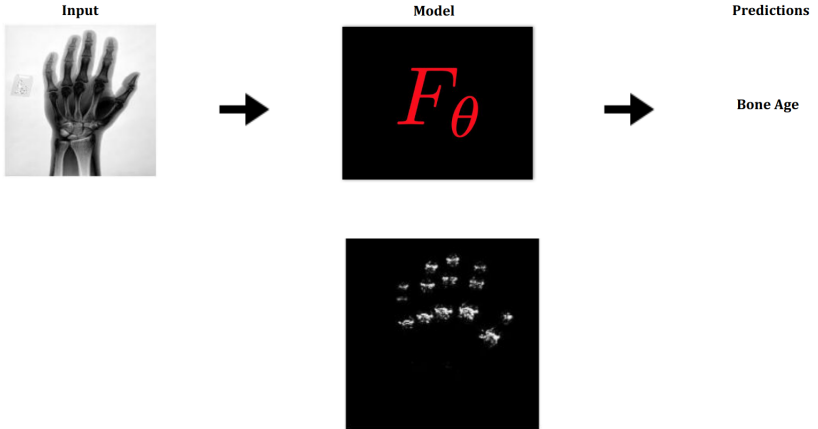
- Different types of variables in the data
 - **Different types of variables** call for different similarity/perturbation functions
 - Gradients may not always be meaningful
- Depending on the task/domain, data could be either **low or high-dimensional**
 - E.g., movie recommendations – user x movie matrix (high dimensional)

- **Feature importance** based explanations
 - Perturbation methods e.g., LIME
 - Saliency maps and other gradient based methods not very meaningful
- **Prototype/example** based explanations
 - Might not always be interpretable
 - E.g., an instance with 100 feature values as prototype
- **Rule-based** explanations
- **Counterfactual** explanations

[Koh and Liang, 2017; Letham et al., 2015; Lakkaraju et al., 2016; Ustun et al., 2019; Rawal and Lakkaraju, 2020]



Source: H. Lakkaraju, J. Adebayo, S. Singh

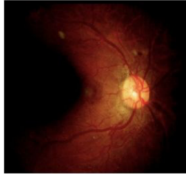


Source: Unknown

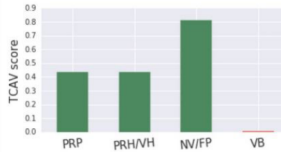


Source: [Sayres et al., 2019]

DR level 4 Retina



TCAV for DR level 4



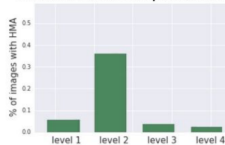
DR level 1 Retina



TCAV for DR level 1



HMA distribution on predicted DR



Source: [Kim et al., 2018]

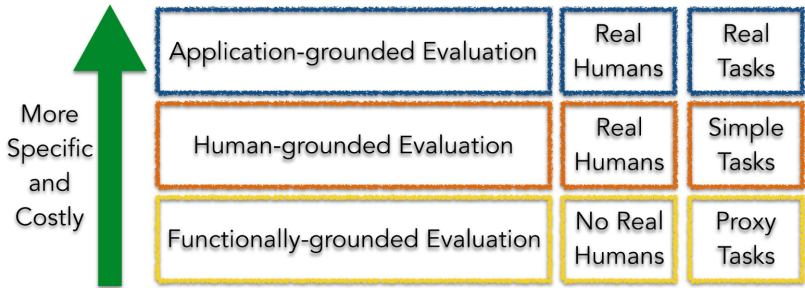
- Lots of **NLP applications** everywhere
 - Translation, social media analysis, hate speech filtering, digital assistants, ...
- **Quickly evolving in major ways**, last few years
 - Word embeddings, ELMo, BERT, GPT-2/3, T5, ...
- Large gap between benchmark numbers and how good methods really are
 - Lots of question answering, classification, textual entailment, etc., are considered "solved"
- **Unique and additional** challenges (that are more general)
 - Domains with discrete/structured/combinatorial inputs ...

- **Discrete** space of inputs
 - E.g. gradients are not directly applicable (or as meaningful)
- Not all combinations of words are well defined
 - They need not be nonsense, ungrammatical, etc
- Difficult to write similarity/perturbation functions
- Format is not fixed: not everything is a classification
 - Structured prediction, text generation, span selection, ...
- Language does not lend itself to "simple explanations"

- [*https://captum.ai/tutorials/Bert_SQUAD_Interpret*](https://captum.ai/tutorials/Bert_SQUAD_Interpret)
- [*https://github.com/PAIR-code/lit*](https://github.com/PAIR-code/lit)
- [*https://allennlp.org/interpret*](https://allennlp.org/interpret)
- [*https://github.com/QData/TextAttack*](https://github.com/QData/TextAttack)
- [*https://github.com/interpretml/interpret-text*](https://github.com/interpretml/interpret-text)
- Influence functions for text [!\[\]\(c8dce68b26731c7aa5915072fc9d68dd_img.jpg\)](#)
- Triggers Code [!\[\]\(76b3245de86167eba9fcdc9cc9f32aa4_img.jpg\)](#)
- Anchors Code [!\[\]\(13db7587f50867332e5bedc6a161739d_img.jpg\)](#)
- LIME Code [!\[\]\(7be5ea91065783fbb69e41ba5d9680f7_img.jpg\)](#)

Explainable Machine Learning

Evaluation of Explanations



Source: H. Lakkaraju, J. Adebayo, S. Singh

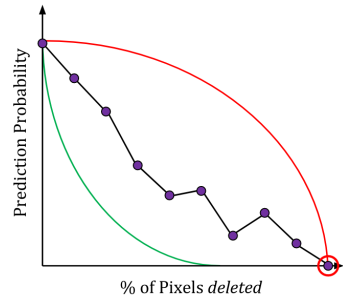
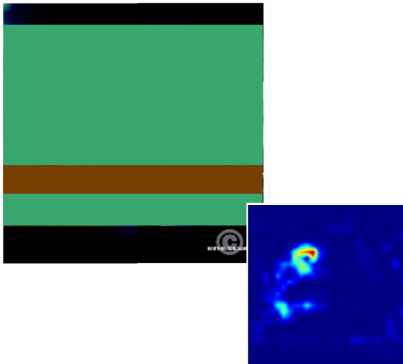
		What are you evaluating?		
		Understand the Behavior	Useful for Debugging	Help make decisions
How we evaluate it?	Application- grounded			
	Human- grounded			
	Functionally- grounded			

Source: H. Lakkaraju, J. Adebayo, S. Singh

Slide source: [Lage et al., 2019]

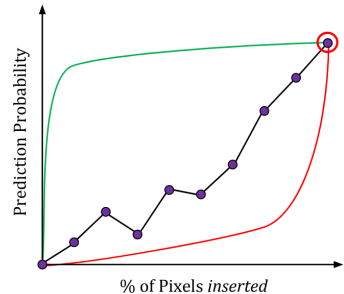
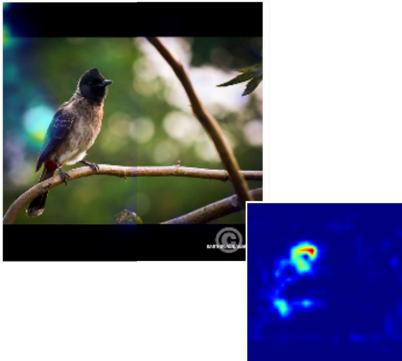
- Understand behaviour
- Useful for debugging
- Help make decisions

Deletion: Remove important features and see what happens...



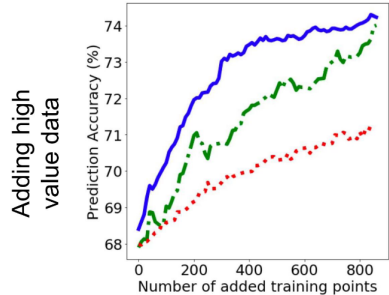
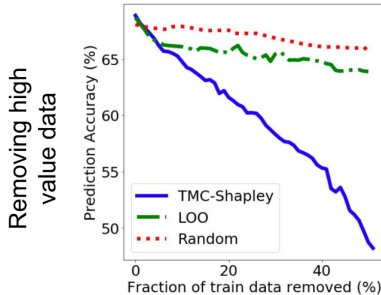
Source: [Qi et al., 2020]

Insertion: Add important features and see what happens...



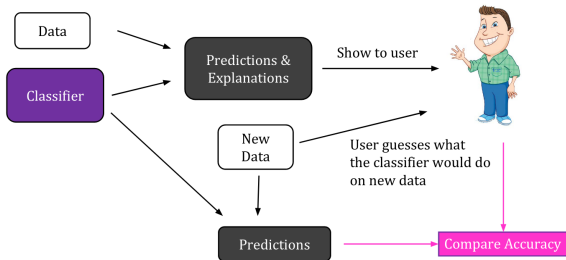
Source: [Qi et al., 2020]

Idea: Add/remove influential training data (order of removal depends on the importance coming from the explanation), see what happens...



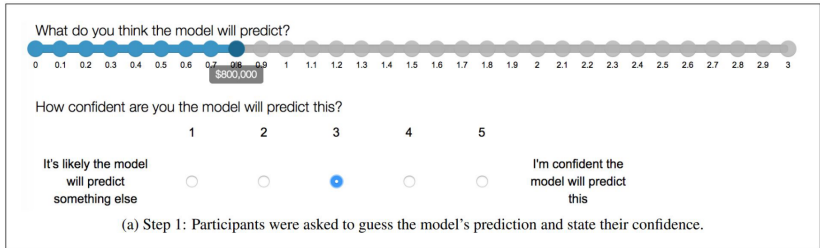
Source: [Ghorbani and Zou, 2019]

Idea: Analyse whether a user is able to predict what a classifier will do on new data given other predictions and explanations



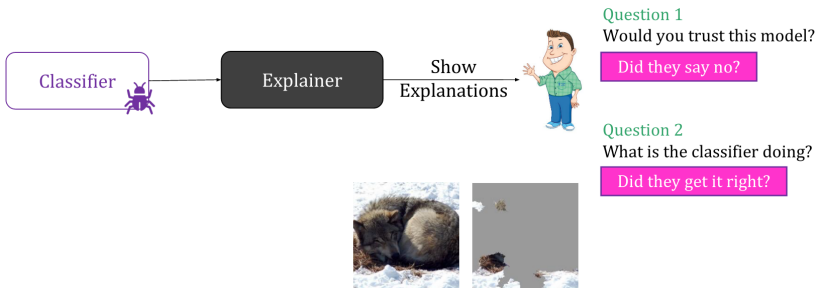
Source: H. Lakkaraju, J. Adebayo, S. Singh

Idea: Analyse whether a user is able to predict what a classifier will do on new data given other predictions and explanations



Source: [Poursabzi-Sangdeh et al., 2021]

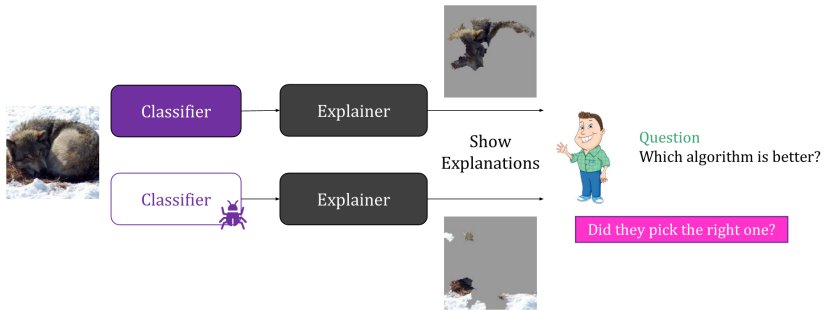
Idea: Create a buggy classifier and analyse whether a user is able to figure out what this bug is



Source: H. Lakkaraju, J. Adebayo, S. Singh

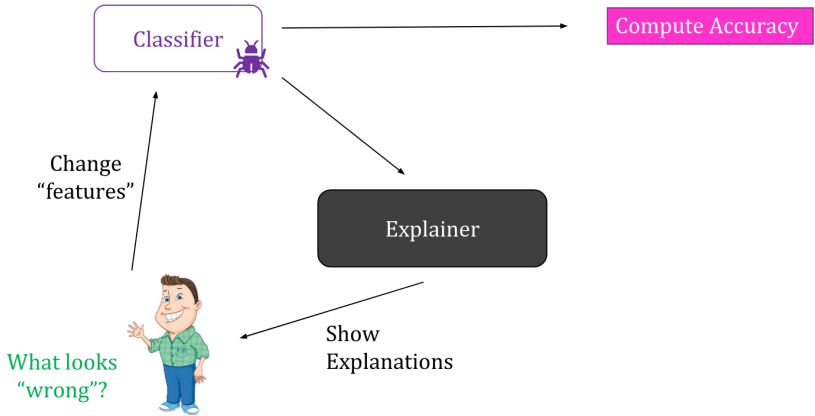
Slide source: [Ribeiro et al., 2016]

Idea: Analyse whether the user, given two explanations, can correctly say (more often) which method is better



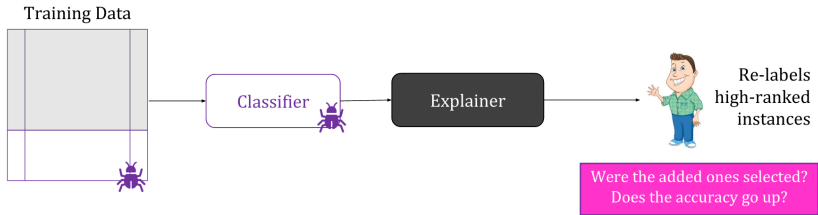
Source: H. Lakkaraju, J. Adebayo, S. Singh

Slide source: [Ribeiro et al., 2016]



Source: H. Lakkaraju, J. Adebayo, S. Singh

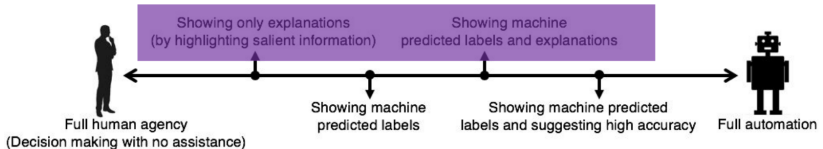
Idea: Add noisy training instances and see whether, given the explanation, a user can select and re-label these successfully



Source: H. Lakkaraju, J. Adebayo, S. Singh

Are explanations useful for making decisions?

- For tasks where the algorithms are not reliable by themselves



Source: H. Lakkaraju, J. Adebayo, S. Singh

Slide source: [Lai and Tan, 2019]

Deception Detection: Identify fake online reviews

- Are humans better deception detectors with explanations?

Note: The highlighted words are important words which machine learning classifiers use to decide if a review is genuine or deceptive. The below scale shows level of importance of each word.



I would not stay at this hotel again. The rooms had a fowl odor. It seemed as though the carpets have never been cleaned. The neighborhood was also less than desirable. The housekeepers seemed to be snooping around while they were cleaning the rooms. I will say that the front desk staff was friendly albeit slightly dimwitted.

Genuine

Deceptive

Source: H. Lakkaraju, J. Adebayo, S. Singh

Slide source: [Lai and Tan, 2019]

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Open projects

Computer Vision Human-Computer Interaction Machine Learning Natural Language Processing

Bachelor Master

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(Guanhua Zhong)

Mental Face Reconstruction by Interactive Morphing
(Florian Strahm)

Image Synthesis for Appearance-based Gaze Estimation
(Mihai Băice)

Multi-duration Saliency Prediction for Mobile User Interfaces
(Mihai Băice)

Visual Scanspath Prediction for Mobile User Interfaces
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AI-assisted Semi-automatic Labelling of Eye Contact Data
(Mihai Băice)

Multi-Step Visual Reasoning in Video Grounded Dialog
(Adrien Abdessattar)

Diagnostic Dataset for Video Dialog based on CLEVRER
(Adrien Abdessattar)

Open Projects

collaborative-ai.org/teaching/open-projects/

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